REQUIREMENT ANALYSIS
Technology Stack

Date	01 JULY 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

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1. Introduction

The **Personalized Networking Assistant (Smart Connect AI)** is developed using a modern technology stack that combines **Artificial Intelligence**, **Natural Language Processing**, **FastAPI**, **Streamlit**, and the **Wikipedia API**. These technologies work together to provide users with an intelligent, fast, and interactive networking assistant for generating conversation starters, verifying knowledge, saving networking sessions, and collecting feedback.

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2. Technology Stack Overview

Layer	Technology Used	Purpose
Programming Language	Python 3.x	Core application development
Frontend	Streamlit	Interactive web application interface
Backend	FastAPI	REST API development and request handling
AI/NLP	Python-based NLP	Event analysis and conversation generation
Knowledge Source	Wikipedia API	Topic verification and information retrieval
HTTP Communication	Requests Library	Communication between frontend and backend
Data Storage	JSON Files	Store networking history and user feedback
Server	Uvicorn	Run FastAPI backend server
IDE	Visual Studio Code	Code development and debugging
Version Control	Git & GitHub	Source code management

3. Programming Language

Python 3.x

Python is the primary programming language used to build the application because of its simplicity, powerful libraries, and strong support for Artificial Intelligence and web development.

Advantages

- Easy to learn and maintain
- Supports AI and NLP libraries
- Fast development
- Large community support
- Cross-platform compatibility

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4. Frontend Technology

Streamlit

Streamlit is used to build the interactive web interface that allows users to interact with the Personalized Networking Assistant.

Responsibilities

- User-friendly dashboard
- Event description input
- Conversation starter display
- Knowledge Assistant interface
- Session saving
- Feedback collection
- Interactive notifications

Advantages

- Rapid UI development
- Responsive design
- Easy integration with Python
- Minimal coding effort

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5. Backend Technology

FastAPI

FastAPI provides REST API services that process requests from the Streamlit frontend.

Responsibilities

- Event analysis
- Generate conversation starters
- Verify knowledge
- Save networking history
- Store user feedback
- API communication

Advantages

- High performance
- Automatic API documentation
- Fast request handling
- Easy integration

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6. Knowledge Source

Wikipedia API

The Wikipedia API provides trusted and verified information for networking topics entered by users.

Functions

- Topic search
- Summary generation
- Information verification
- Reliable knowledge retrieval

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7. Data Storage

JSON Files

The application stores data locally using JSON files.

history. json

Stores:

- Event Description

- Generated Topics
- Date & Time

feedback. json

Stores:

- User Feedback
- Date & Time

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8. API Communication

Requests Library

The Requests library enables communication between the Streamlit frontend and the FastAPI backend.

Functions

- GET Requests
- Receive JSON Responses
- Handle API errors
- Transfer user data

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9. Development Tools

Tool	Purpose
Visual Studio Code	Code Editor
Git	Version Control
GitHub	Source Code Repository
Uvicorn	FastAPI Server
Streamlit	Frontend Execution
FastAPI	Backend Execution

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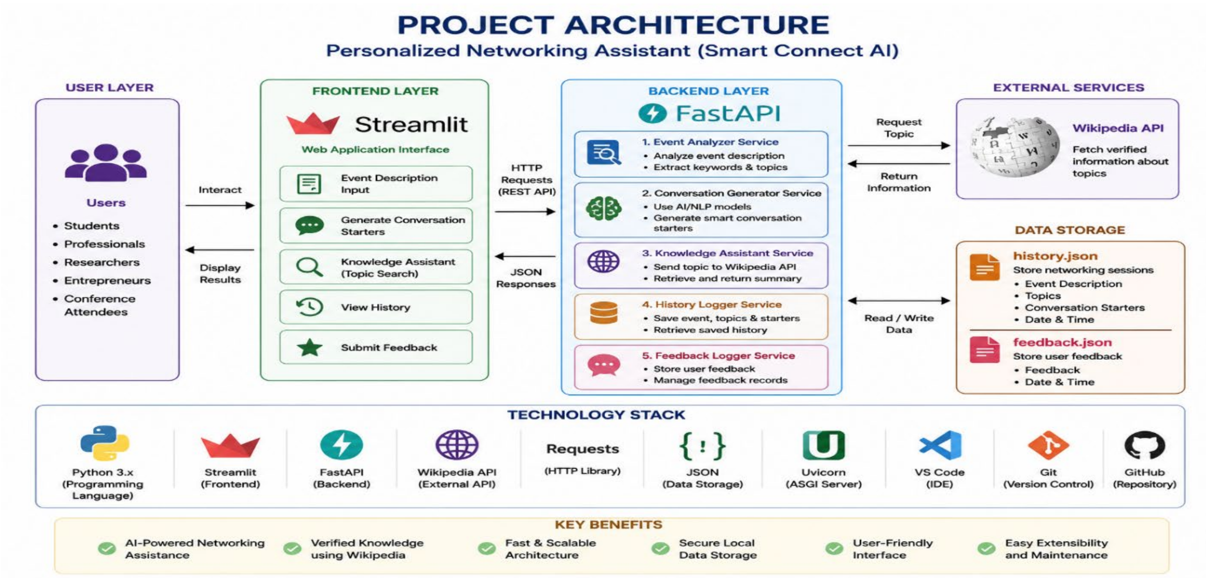
10. Project Architecture



11. Technology Benefits

The selected technology stack provides:

- Fast application performance
- Simple and responsive interface
- Easy deployment
- Modular architecture
- AI-powered networking support
- Reliable knowledge verification
- Secure local data storage
- Easy future enhancements

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12. Conclusion

The **Personalized Networking Assistant (Smart Connect AI)** utilizes a modern and efficient technology stack consisting of **Python**, **FastAPI**, **Streamlit**, **Wikipedia API**, **Requests**, and **JSON Storage**. This combination enables intelligent event analysis, AI-powered conversation generation, verified knowledge retrieval, session management, and user feedback collection through a responsive and user-friendly web application. The chosen technologies ensure scalability, maintainability, and high performance, making the solution suitable for students, professionals, researchers, and entrepreneurs attending networking events.